PHYSICS 253A, 253B, AND 253C

Quantum Field Theory

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ABSTRACT: A source-faithful textbook edition of the Physics 253abc lecture notes and recorded lectures. This pilot edition contains Chapter 1.

Contents

Part I

Physics 253a: Quantum Field Theory I

1 Basic Generalities of Quantum Field Theory

1.1 What is quantum field theory?

What is quantum field theory? This question is very easy to answer.

A rough answer would be: it's basically quantum mechanics, but with relativistic locality. And I haven't explained what that means.

So, in fact, this is actually a confusing question even for the vast majority of physicists, because I would say that there are really two conceptually slightly different things that are both called quantum field theory, that are really conceptually distinct.

I want to discuss both in this course. One type of quantum field theory is what I would call local, or sometimes referred to as UV complete, and the other is what's known as effective field theory.

So what are these things? It's a little easier to state what a local QFT means.

First of all, a local QFT is equipped with Poincaré symmetry. Poincaré symmetry consists of translations in spacetime together with Lorentz transformations.

This kind of fundamental theory, first of all, is a quantum-mechanical system. Some folks might have the slight misconception that quantum field theory, going beyond quantum mechanics, is a generalization of quantum mechanics: no, quantum field theory is not a generalization of quantum mechanics. Quantum field theory is a specialization of quantum mechanics.

Okay, so we are going to allow quantum mechanics with infinitely many degrees of freedom.

The only requirement, well, not the only requirement, but one requirement on top of the kind of quantum mechanics you learn in your undergraduate quantum-mechanics class is that we must also have Poincaré symmetry.

In a quantum-mechanical system, a symmetry will be realized as a unitary operator that acts on Hilbert space, and an operator will transform by conjugation with the action of this unitary operator. So that will also be the case here.

I am giving a very broad overview for now. All of these words will be put in the form of precise equations later.

But that's not all. We'll see that for a local QFT there also have to be local operators, sometimes referred to as field operators.

Well, we'll say what they are. Importantly, these field operators have to obey some important properties, including, particularly, microcausality.

Basically, this is a precise quantum-mechanical statement that captures the idea that signals cannot propagate faster than the speed of light in a relativistic theory.

Okay, now, what is effective field theory?

Well, it's not really possible for me to explain in detail, without setting up some technical tools, what effective field theories are. At this point, I'll just say a few general facts about it.

Typically, it is defined only perturbatively in some coupling.

So, by contrast, there are plenty of examples of quantum field theories that are completely well-defined at strong coupling. These effective field theories are typically only defined as a formal perturbative series in the coupling constant, and it may not have a well-defined limit. In particular, it may or may not converge.

And it's going to be based on a path integral, a functional integral over the space of fields.

So you can also use that to define a local fundamental theory, but very often, depending on the details of the theory, it may or may not make sense at the nonperturbative level.

An effective field theory typically, by design, captures long-distance or low-energy observables, and not necessarily beyond that.

So, let me just say very roughly, for a moment, that we'll have some field operator. Let's call it $\hat{\phi}$, which will be associated with a spacetime point. Let's say it depends on time coordinate t and space coordinate x.

To say that you have an operator associated to a point, by itself, is not a very meaningful statement. I can parametrize $\hat{\phi}$ however I want.

Now, there are two key criteria, which I'll write as equations later, but I'll just say them now, that say what it means to be local. First of all, under a Poincaré transformation, if you move a point to another point in spacetime by a Poincaré-symmetry action, the corresponding unitary operator acts on $\hat{\phi}$ by moving it to $\hat{\phi}$ at the other point.

Also, on top of that, these operators have to obey microcausality, in the sense that you have $\hat{\phi}$ at one point in spacetime and another at another point in spacetime. If these two points are spacelike separated and causally disconnected, then the two operators commute. So that's kind of microcausality. These two are the essential criteria for what it means for the operator to be local. This is a concept that's rather alien if you only know about undergraduate quantum mechanics, but that will be central to our study of QFT.

All right, so what are some examples? Let me write down some examples of local QFT versus effective field theory, some of which we'll discuss in this course, and some in the next semester's course.

So, there's the famous ϕ^4 theory. This is the theory of a scalar field with quartic interaction. We'll introduce this in a few weeks, actually.

But now, not in four-dimensional spacetime, but rather in D. This is the spacetime dimension. Whenever I use capital D, I always mean the spacetime dimension here, including time and space. So, if you construct such a theory in two or three dimensions of Minkowski spacetime, so either $1 + 1$D or $2 + 1$D, then that would be a completely fine local QFT.

But on the other hand, ϕ^4 theory in four dimensions is only sensible, only makes sense, at the level of effective field theory. All right.

Now, some of your favorite theories that you might want to study are, unfortunately, actually not known to be well-defined at the nonperturbative level as local QFTs. QED is one example.

Neither is the Standard Model, so all of that belongs to the effective field theory. In fact, this might make you question whether there exist any local QFTs at all in four dimensions, and, surprisingly, yes.

But it was not obvious why. The theories I mentioned are things like Yang–Mills theory, and also QCD, provided you don't have too many flavors, of course.

And there are other examples of local quantum field theory that arise in different contexts. So far, things like QED, the Standard Model, or Yang–Mills theory all arise in particle physics. Quantum field theories have many more applications that go far beyond particle physics. For example, we'll see the statistical Ising model near criticality.

This is something that, maybe not so much for this semester, but in the next semester will be discussed.

The statistical Ising model is captured by the QFT. Now, in this context, I will be talking about observable quantities in statistical equilibrium, so there's no time evolution. But, in fact, it will turn out that the statistical averages have all the good properties.

There are other examples of effective field theory. If you try to quantize GR, you get general relativity as an effective field theory. There's also something called chiral perturbation theory. It captures the low-energy interactions of pions in QCD, which is also a famous example.

Any questions? Let me just mention that, you know, there are these historical terms of renormalizability and non-renormalizability that really refer to, I would say, the perturbative theories, in a sense that we'll not quite discuss in detail now, but we'll come to this later.

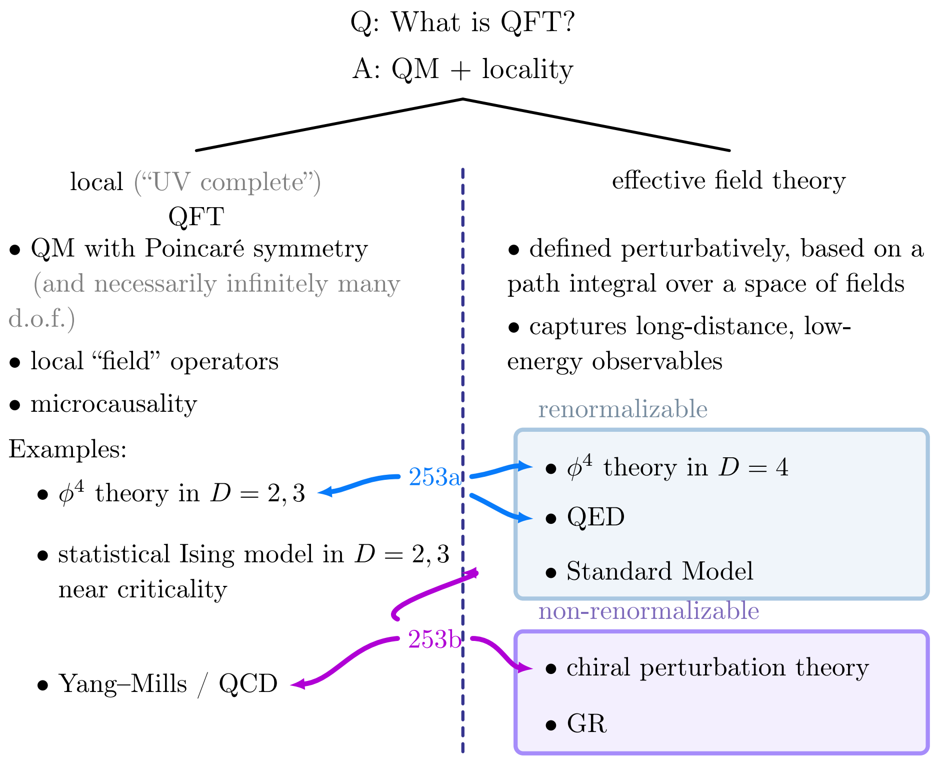

Figure 1. The opening map of local QFT and effective field theory. The colored arrows retain the source allocation between Physics 253a and Physics 253b.

1.2 Plan of Physics 253a

So now let me describe the plan of all these lectures. The plan of the course is the following.

I'm going to begin with the more familiar subject of just plain old nonrelativistic quantum mechanics with a few degrees of freedom. But we're going to study quantum mechanics in a language that's perhaps unfamiliar from your undergraduate quantum mechanics course, but will be the standard technical tool in the study of quantum field theory. So while in your undergraduate quantum mechanics class you probably learned it in the language of the Hamiltonian, we're going to be considering the Lagrangian formulation of quantum mechanics.

Now we'll come to this in more detail in the next lecture. But, as you probably know, in classical mechanics the Lagrangian and the Hamiltonian formulations are more or less equivalent, right? So you can define a mechanical system with its Lagrangian, or action, and you can derive the Euler–Lagrange equation, which is equivalent to the extremization principle of the action, and finally you write the equation of motion.

Alternatively, you can describe the classical state of our system in terms of a point in phase space, captured by the coordinates and momenta. And then you can write a Hamiltonian, which is related to the Lagrangian by a Legendre transformation, and then you have the Hamiltonian equations of motion, which are equivalent.

Presumably, usually when you learn this in classical mechanics, you're not very impressed, because these are just trivial rewritings of one another. So why bother?

But in a quantum-mechanical context, you probably noticed, in the Hamiltonian language, that the Hamiltonian, rather than being just a classical function of coordinates and momenta, is promoted to a quantum operator that acts on Hilbert space, whose eigenvalues are the energy spectrum of your system. So that's an essential object in quantum mechanics, whereas the Lagrangian does not then have a prominent role, right? In fact, it's not even clear whether the Lagrangian is, you know, rigorously defined or meaningful for a quantum-mechanical system.

Okay. In fact, really the only way I know to formulate quantum mechanics, I mean formally, through a Lagrangian is through the path integral. So we'll spend the time discussing the path-integral formulation of quantum mechanics. And, importantly, we need to encounter the concept of regularization, which will be very standard in quantum field theory. But, as we'll see, it is already there in good old quantum mechanics with just a few degrees of freedom. Then we'll discuss perturbation theory in the language of the path integral in quantum mechanics, from which we'll also derive things such as Feynman diagrams. We'll see that Feynman diagrams, in principle, have nothing to do with quantum field theory. They have everything to do with the perturbative expansion of the path integral.

And we'll also discuss renormalization and the so-called counterterms, which, as you'll see, also have nothing to do with quantum field theory, and have everything to do with a path-integral formulation of quantum mechanics.

So, again, we'll develop these basic concepts along the way, and then we'll move to field theory. So the second part of this semester, well, not the second part, but in a few weeks we'll discuss relativistic particles and fields. So we'll discuss things like scalar field theory, like ϕ^4 theory.

We'll discuss the notion of Green functions in quantum field theory, the notion of asymptotic states in a scattering problem. These are kind of more particle-physics-oriented concepts.

And we'll discuss the so-called scattering matrix, or S-matrix, which is the central observable of a large part of quantum field theories, and how they're related to the Green function by the so-called LSZ reduction. Yeah.

And through that we also will understand why it is that the Feynman diagrams can be used to calculate scattering amplitudes.

And then, in the third part of this semester, there's a big set of topics. We will discuss particles and fields with spin.

Now, I would like to always distinguish the concepts of particles and fields, because they are really, really different things. By a field, I generally mean the local field operator, in this kind of handwriting. And particles are particles: if it smells like a particle, looks like a particle, it will be a particle. Particles and fields are really very different concepts.

And the particle can carry spin, and fields can carry spin. But the spin of the particle and the spin of the field have nothing to do with each other. We'll see what they mean, and we'll also discuss the classification of relativistic particles: what are the particles that there can be? We'll understand why it is that you can have spin-one-half fermions and not, say, spin-one-third particles in four dimensions.

And then we'll discuss things like fermions and gauge bosons in preparation for QED.

And then there will be some standard applications that we'll come to at the end of this sequence: the magnetic moment of the electron and things like that.

The source plan is:

1. Lagrangian formulation of quantum mechanics:
   - path integral and regularization;
   - perturbation theory and Feynman diagrams;
   - renormalization and counter terms.
2. Relativistic particles and fields:
   - ϕ^4 theory;
   - Green functions;
   - asymptotic states;
   - S-matrix and LSZ reduction.
3. Particles and fields with spin:
   - classification of relativistic particles;
   - fermions and gauge bosons;
   - QED;
   - applications to the electron g-factor and the Lamb shift.

Canonical quantization will be somewhere here. We'll discuss the quantization of fields in preparation before setting up a scalar field theory.

1.3 Relativistic particles without fields

It's okay. But in today's lecture, I'd like to at least give not only an overview, but some motivation for why I'm going to talk about fields.

In fact, the first question is: why do particle physicists need to know field theory? Why isn't it enough to just talk about particles? That's a very good question you should ask yourself. It's not obvious why. I'll attempt to explain why that is. So the question is: why do we need field theory to describe the quantum mechanics of particles with relativistic symmetry? Okay, this is not obvious.

Let's try. As in any quantum-mechanical system, you have a Hilbert space of states. Every vector in the Hilbert space corresponds to a physical state in the quantum-mechanical system. There's probably going to be a vacuum state. Let's call it $|\Omega\rangle$.

And we're going to assume, and we'll see that this is a reasonable assumption, that the vacuum state is invariant under Poincaré symmetry. If it's not invariant, then if you perform a Poincaré transformation, you get a different vacuum state, so that sounds unreasonable.

And if, in particular, one of the generators of Poincaré symmetry is the time-translation generator, which is the Hamiltonian, that will tell us that the vacuum state should have zero energy:

$$H|\Omega\rangle = 0. \tag{1.1}$$

And on top of that, we can have some particles. So let's say maybe we have some single-particle state. A relativistic particle will be labeled, and you can find a basis for the particle such that the basis states are labeled by spatial momentum. We can write $|\mathbf{p}\rangle$ for the particle. So I should state my convention.

I will always write the Minkowski energy-momentum vector, say in three-plus-one dimensions, as p^μ. Its time component, the energy, will be p^0, and the spatial momentum will be $\mathbf{p}$. Unless stated otherwise, I'm generally going to work in arbitrary spacetime dimension. So μ will be zero, one, two, and beyond. If you're a particle physicist, you might wonder why you would care about a dimension other than four. Well, here's an example: if you study the Ising model, it describes an actual object in solid-state physics.

So I'm going to keep the spacetime dimension D general, but often, for now, you can think of $\mathbf{p}$ as three-dimensional. Sorry, D will be the spacetime dimension, so μ goes from zero to $D-1$.

And so, if you have a relativistic particle, it typically is going to have a definite mass, and you probably know the energy of the particle. It is supposed to obey the relativistic dispersion relation.

So if the vacuum energy is zero, then the Hamiltonian acting on a one-particle state, for the particle of definite momentum, definite spatial momentum, is going to give you the energy $\sqrt{\mathbf{p}^2 + m^2}$, where m is the mass of the particle:

$$H|\mathbf{p}\rangle = \sqrt{\mathbf{p}^2 + m^2}\,|\mathbf{p}\rangle. \tag{1.2}$$

Okay, so far, nothing, no big deal. Now you can also imagine multiparticle states. So let's imagine we have an n-particle state consisting of n particles labeled by their spatial momenta, $\mathbf{p}_1, \mathbf{p}_2$, all the way to $\mathbf{p}_n$. They could be identical particles.

And I'm going to assume for now that the particles are non-interacting. So you can say we have a very simple physical system whose Hilbert space consists of states with arbitrary numbers of particles, but these particles are not interacting. Let's say they all have the same mass, and perhaps they're identical particles. Then the Hamiltonian is going to act on the state $|\mathbf{p}_1, \ldots, \mathbf{p}_n\rangle$, and the energy will just be the sum of the energy of each one of those particles.

$$H|\mathbf{p}_1, \ldots, \mathbf{p}_n\rangle = \left(\sum_{i=1}^{n} \sqrt{\mathbf{p}_i^2 + m^2} \right) |\mathbf{p}_1, \ldots, \mathbf{p}_n\rangle. \tag{1.3}$$

Those are all the states, and we have already specified how the Hamiltonian acts on every single state. We're all set. We have solved this one kind of system, and there's nothing more to do.

A student asks how the particles or states are created. Yes. I don't really care. I've just declared the states are there. All right, I mean, the system is non-interacting, and that's my system. Later we'll study interacting systems, and you can create particles with scattering, but I don't have to say how I created the state. It exists. Now, that's the system I'm going to study for the moment. It's a very simple system.

And you're always allowed to consider linear superpositions of the states. But when I say non-interacting, I just mean that the energy of any particular state is the sum of the energies of these particles. There's no potential contribution due to any mutual interaction.

Maybe a rather boring one, but the reason I'm talking about this is just to explain that you can easily construct a Lorentz-invariant quantum-mechanical system without ever talking about fields. Now, here it is.

There's a question about, you know, creation, how to create the state. I said, you know, that's not something I need to discuss, but we can, if you wish, define creation and annihilation operators in the following way at this point. So we can introduce particle creation and annihilation operators.

We usually call these $a^\dagger$ and a, which are also labeled by the momentum of the particle. I'm going to write that label as a subscript $\mathbf{p}$. Note that to specify the momentum of the particle I just need to specify the spatial momentum, and the energy is always, in this case, determined by the dispersion relation. So I don't have to say in addition what the energy is.

Notation. The handwritten creator is denoted by $a_{\mathbf{p}}^{+}$, with $a_{\mathbf{p}}^{+} \equiv a_{\mathbf{p}}^{\dagger}$. So I'm going to define the creation operator by how it acts on these states. Well, the way that you can define that is more or less obvious. So, for example, the one-particle state should be the creation operator $a_{\mathbf{p}}^{\dagger}$ acting on the vacuum.

And if I would like to say a two-particle state, you just act with two of these creation operators on the vacuum, and so forth:

$$|\mathbf{p}\rangle = a^+_{\mathbf{p}}|\Omega\rangle, \qquad |\mathbf{p}_1, \mathbf{p}_2\rangle = a^+_{\mathbf{p}_1} a^+_{\mathbf{p}_2}|\Omega\rangle, \qquad \text{etc.} \tag{1.4}$$

Now, I haven't so far specified the normalization of this basis of multiparticle states. Let me specify that now. It's important for me to specify the commutation relation of a and $a^\dagger$. By the way, of course, I should have said that these commute.

And then I have to tell you the commutation relation of these creation and annihilation operators. For example, the two creation operators commute, and so do the two annihilation operators.

There should be a nontrivial commutation relation between the annihilation operator and the creation operator. So we do $a_{\mathbf{p}}$ and $a^\dagger_{\mathbf{p}'}$, and this should be nonzero only if the particle momenta coincide. And so here let me choose the normalization convention. This is the delta function, so delta in $D-1$ spatial dimensions, as shown below.

As you can see, if I basically take the overlap $\langle \mathbf{p}|\mathbf{p}'\rangle$, the one-particle state with momentum $\mathbf{p}'$ can be written as $a^\dagger_{\mathbf{p}'}$ acting on the vacuum. The vacuum bra comes with an $a_{\mathbf{p}}$.

There is a vacuum bra, actually, with the Hermitian conjugate, which gives an $a_{\mathbf{p}}$. And then I use the commutation relation to replace the product by the commutator. So the result is the delta function shown below.

So I'm just saying that I'm choosing a normalization for my one-particle state such that the overlap between a one-particle state and another is the delta function shown below.

$$[a, a] = [a^+, a^+] = 0, \qquad [a_{\mathbf{p}}, a^+_{\mathbf{p}'}] = \delta^{(D-1)}(\mathbf{p} - \mathbf{p}'), \qquad \langle \mathbf{p} \mid \mathbf{p}'\rangle = \delta^{(D-1)}(\mathbf{p} - \mathbf{p}'). \tag{1.5}$$

A student asks why, sometimes, in some places, this delta function appears with an extra factor. That is a matter of normalization convention, right? I'm free to rescale a and $a^\dagger$ by a factor of two, and I would just change my normalization convention. Now, you see, this one-particle state corresponds to a plane wave of the particle, which, strictly speaking, is not a normalizable state anyway. If you want to form a normalizable state, you have to take a wave packet. For example, you can take some state like that, you know, by integrating these states against a function. The overall factor is another convention. I'm allowed, for example, also to rescale this state by a factor of the energy to the one-third power, or whatever.

A student asks what the annihilation operator does. The annihilation operator should reduce the particle number by one, just like a creation operator should increase the particle number by one. So the vacuum has zero particles, and that's part of the definition.

Any other questions? A student asks whether we can choose the energy of the vacuum state however we want. Right, so that's a good question. I stated this as following from our assumption that the vacuum is Poincaré invariant. Now, of course, you could trivially redefine your Hamiltonian by adding a constant. You're allowed to do that. But I'm going to choose the vacuum energy to be zero. It's a natural assumption.

We would like to say that there is a vacuum, though it need not be unique. I would assume that this vacuum state is symmetry-invariant, and we'll see shortly, in a bit, that

H is one of the generators of our symmetry. And to say the state is invariant under the symmetry means all the generators should annihilate it, as in any continuous global symmetry.

All right. So, given these creation and annihilation operators, well, why do I do this? Well, we can then express the Hamiltonian directly in terms of these creation and annihilation operators without having to say how it acts on the basis of states.

So you can easily verify that, in terms of a and $a^\dagger$, we can write the Hamiltonian as the integral below.

$$H = \int d^{D-1}\mathbf{p}\,\sqrt{\mathbf{p}^2 + m^2}\,a^+_{\mathbf{p}} a_{\mathbf{p}}. \tag{1.6}$$

The operator $a^\dagger_{\mathbf{p}} a_{\mathbf{p}}$ is the occupation number for the particles. So, I'll let you verify for yourself using the commutation relation that this Hamiltonian agrees with the definition and produces the expected energy.

That's much less obvious. We'll see that there are massless particles, and we can also study what we call a massless theory. And for massless particles, it's not obvious why you want to demand the number of particles to be finite, and that is a very subtle question. We may not even be able to get to it this semester, but it's definitely tied to the issue of infrared divergences in quantum electrodynamics.

All right. So here it is. We have a well-defined system, with its Hilbert space and Hamiltonian. And by design, this is a quantum-mechanical system of free relativistic particles.

So if you're just interested in describing quantum mechanics of free particles, free in the sense that they're not interacting with one another, but then we're done. And so then you don't have to take this course; you can go home. But the thing is, we would like to be able to describe interactions. So what about interacting particles?

Well, you might say that that's not a big deal. So, you know, I can take, so let's call this H_{free}; let's call this H_0, for the free theory. So in quantum mechanics we can just pick whatever Hermitian operator we would like, so you can add some interaction terms.

And let's try that. Okay, so, for example, you might even include interaction terms in the Hamiltonian that allow changing particle numbers. So, for example, you can have operators like $a^\dagger a^\dagger a$, with some momentum assignments. Sorry, let's say like this, or $aaa^\dagger$. So this guy, for example, can take this one-particle state, annihilate the one particle, and create two particles. So this operator would act on the one-particle state and give you a two-particle state. This internal kind of interaction will change the number of particles.

$$H = H_0 + \underbrace{H_{\text{int}}}_{a^+a^+a,\; aaa^+,\; \dots}. \tag{1.7}$$

Hermiticity. Momentum kernels and Hermitian-conjugate terms are understood so that H_{int} is Hermitian. Okay. So we can do that. So what's the big deal?

The issue is the following. So in principle you could have done that, but it's quite difficult to introduce interactions in this manner in a way that preserves relativistic symmetry.

Okay, so the issue is that it is possible, but not easy, not easy to find H_{int} that respects relativistic symmetry.

1.4 Local disturbances and causality

Okay, so let me draw a schematic picture of space and time, with space labeled by some spatial coordinate $\mathbf{x}$, and time. If we have some event or some disturbance of the system that occurs at one point in space, say at time zero at some location in space, let's say here's the light cone. A light ray travels in this picture. I'm only drawing one spatial direction, right? I'm also working in units where the speed of light is equal to one, so light will travel along rays at forty-five-degree angles, left and right.

A trajectory like this would be timelike; it would correspond to propagation of a signal slower than the speed of light. Whereas a trajectory like this, outside of the light cone, would be superluminal, and that's not supposed to be allowed in a relativistic theory. So superluminal propagation should be forbidden.

So I'm not allowed to say that I can, you know, disturb the system here, there's a red point here, and measure some response at that point in spacetime. That's not possible. If I disturb the system over here, I should only have a response within the light cone. That's just a constraint from relativistic causality, and any relativistic theory should have this property.

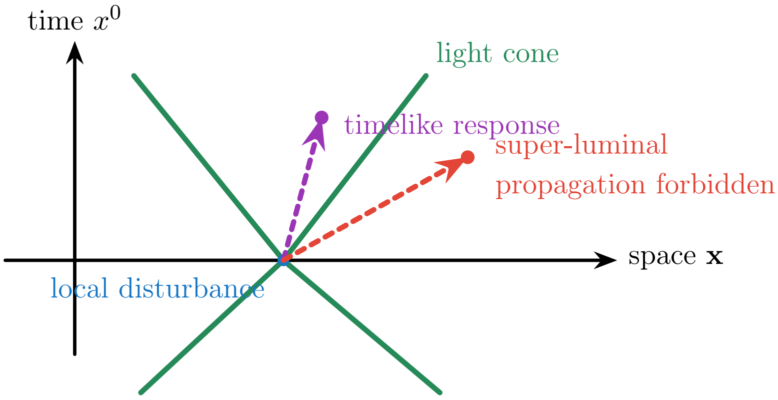

Figure 2. A local disturbance may produce a timelike response inside its light cone. A response at spacelike separation would be super-luminal.

But you see, this kind of property is not built in in the form of the Hamiltonian formulation. Nothing tells you the speed limit. You know, the influence will be a priori instantaneous. In fact, this is not a very easy exercise, but you can try; you're welcome to. Try to include some interaction terms and see if you can arrange them such that there will be no superluminal propagation of signals. And that's actually not an easy thing.

But let me quantify this a bit better in the language of these local field operators.

So in order to actually describe this kind of local disturbance of the system in a way that is mathematically meaningful, suppose you change the state of the system by some operation localized at a point in space, at an instant in time. In a relativistic quantum-mechanical system, you might expect that local disturbance to be represented by some kind of field operator.

And then when I say the word field, what does it mean? All I mean is that the operator depends on some spacetime point x. So that is to say, we have some operator. This operator, in the usual sense of quantum mechanics, acts on the Hilbert space, and we can take it to be Hermitian, if you wish. And, you know, its expectation value is supposed to be some physical observable, in some sense.

But unlike the operators perhaps you are familiar with, what we have here is actually an entire family of operators labeled by some parameter. So I should emphasize that in my notation here. Unlike probably your undergraduate quantum mechanics course, where x and p are operators, here when I write x in this context, they're just parameters; they're not operators. The operator is $\hat{\phi}$.

All right. So this x^μ, which is some spacetime point, specified by the time x^0 and the spatial coordinate $\mathbf{x}$, this is just a label. This is just a label, a spacetime point. And I want to say that associated with that spacetime point there is some operator $\hat{\phi}$ that represents a disturbance of the system.

1.5 Poincaré covariance

Now, this field operator should obey some properties. We said these in words earlier, but let me now say this more precisely.

If I look at $\hat{\phi}(x)$ and $\hat{\phi}(x')$, at different spacetime points x and x', these two operators are supposed to be related by a Poincaré symmetry that sends x to x'.

So, you know, given two spacetime points, and because we're in Minkowski spacetime, I can always find some Poincaré transformation that takes x to x'. This could be just a translation; it could be a different Lorentz transformation, whatever. Of course it's highly nonunique.

You'd expect that this symmetry taking x to x' should also take $\hat{\phi}(x)$ to $\hat{\phi}(x')$. This is a reasonable requirement if you want to say that $\hat{\phi}(x)$ is indeed tied to that spacetime point in some sense.

So, consider a transformation that takes x to x':

$$x^\mu \rightsquigarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu. \tag{1.8}$$

This says that you can shift x by some constant spacetime vector a^μ.

You can also rotate or boost, which is a Lorentz transformation $\Lambda^\mu{}_\nu$.

The transformation matrix $\Lambda^\mu{}_\nu$ is supposed to preserve the Lorentzian metric, so

$$\eta^{\alpha\beta}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta = \eta^{\mu\nu}, \qquad \eta^{\mu\nu} = \mathrm{diag}\,(-1, 1, \ldots, 1)\,, \tag{1.9}$$

where we're using the mostly-positive signature for the Lorentzian metric in D dimensions.

That's the usual Poincaré transformation from special relativity. So now, probably in your freshman mechanics class, you learned Poincaré symmetry as a transformation of spacetime points. But then in quantum mechanics, you learned that a symmetry is supposed to be represented by a unitary operator.

Here we're doing quantum mechanics, so this Poincaré symmetry should also be represented by an actual operator. So the symmetry acts on a state in Hilbert space. How does this symmetry act on the state in Hilbert space? Well, the symmetry is represented by a unitary operator. I'll call it U.

It depends in some way on Λ. The transformation is specified by the Lorentz transformation matrix Λ and the translation by the amount a. So that symmetry will be realized by some unitary operator $U(\Lambda, a)$ that acts on the Hilbert space.

In particular, we need to say how U acts on a field operator.

The state gets transformed by U. But if I have an operator, the operator gets transformed by conjugation, by a similarity transformation with that unitary operator. So, for example, $U\hat{\phi}(x)U^{-1}$.

$$\hat{\phi}(\Lambda x + a) = U(\Lambda, a)\, \hat{\phi}(x)\, \big(U(\Lambda, a)\big)^{-1}. \tag{1.10}$$

So that's the similarity-transformed operator, and our requirement is that it should be equal to the field operator $\hat{\phi}$ at the new point $\Lambda x + a$. I'm using shorthand notation, with the spacetime indices implicit.

Okay, so that's the statement that $\hat{\phi}(x)$ and $\hat{\phi}(x')$, with $x' = \Lambda x + a$, are related by the symmetry. Any questions so far?

A student asks whether the displayed law assumes a scalar field. We will consider that later on. That's a very good point. For now, I'm assuming that this is a scalar field operator.

In fact, there can be many different kinds of scalar operators. I should say that whether the field operator is scalar has nothing to do with what kind of particles we have in the theory. You can have a theory with fermions or gauge bosons where you still have scalar field operators.

So, okay, now what do we know about $U(\Lambda, a)$? Well, so far I haven't told you much. I just said that there should be some operator depending on Λ and a, but exactly how? Well, one way to study a continuous symmetry is to view a finite symmetry transformation as a composition of infinitesimal transformations. It will be convenient to begin by studying the infinitesimal version of the symmetry. So, in fact, in this case it suffices to focus on that. But let us just consider the infinitesimal version of this. We can take $\Lambda^\mu{}_\nu$ to be close to the identity transformation plus a small modification $\omega^\mu{}_\nu$, and a^μ is also small, so $a^\mu = \epsilon^\mu$. You can verify the constraint on this ω for yourself.

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \qquad a^\mu = \epsilon^\mu. \tag{1.11}$$

But, for example, take the condition I wrote above. If you plug in a Λ that is a Kronecker delta plus ω and expand infinitesimally to first order in ω, you'll find that $\omega_{\mu\nu}$, with one index defined by lowering with $\eta_{\mu\nu}$, is antisymmetric with respect to exchanging its indices: $\omega_{\mu\nu} = -\omega_{\nu\mu}$.

This unitary operator U will be very close to the identity. In fact, we can expand to first order in ω and ϵ as the identity operator plus some anti-Hermitian operator. As always, a unitary operator close to the identity can be written as one plus an anti-Hermitian operator. And this anti-Hermitian operator can be written in the following form. There's a term proportional to ϵ^μ, and then there's another that multiplies $\omega_{\mu\nu}$, which I said with lower indices is antisymmetric. The factor of one half here accounts for that antisymmetry, and $\hat{P}$ and $\hat{J}$ are defined by this expansion. So this is just a definition. All I'm saying here is that for this infinitesimal Poincaré transformation, the associated unitary operator can be expanded as the identity plus an anti-Hermitian operator. If I pull out the factor i, then the $\hat{P}$'s and $\hat{J}$'s are going to be Hermitian operators.

They have some names. So this $\hat{P}_\mu$ is the so-called energy-momentum operator.

And the $\hat{J}^{\mu\nu}$ is the operator corresponding to the Lorentz boosts and the angular momentum.

$$U(\Lambda, a) = 1 - i\epsilon^\mu \hat{P}_\mu + \frac{i}{2}\omega_{\mu\nu}\hat{J}^{\mu\nu}. \tag{1.12}$$

Now I'm not going to derive the commutators of these P's and J's, using the property that the U's compose according to the rule of Poincaré transformations.

In particular, $U(\Lambda', a')$, let's say, composed with $U(\Lambda, a)$. Well, this sends x to $\Lambda x + a$, and this one does it again with Λ' and a'. This gives $U(\Lambda'\Lambda, \Lambda' a + a')$, multiplying the matrices.

$$U(\Lambda', a')U(\Lambda, a) = U(\Lambda'\Lambda, \Lambda' a + a'). \tag{1.13}$$

I will let you think for yourself how to derive from this the commutation relations of the P's and J's. The point is that they are all dictated by the requirement that these U's have an interpretation as the unitary operators that generate Poincaré transformations.

$$\begin{aligned} [P, P] &= 0, \\ [P^\mu, J^{\rho\sigma}] &= -i\big(\eta^{\mu\rho}P^\sigma - (\rho \leftrightarrow \sigma)\big), \\ [J^{\mu\nu}, J^{\rho\sigma}] &= -i\big(\eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - (\rho \leftrightarrow \sigma)\big). \end{aligned} \tag{1.14}$$

Well, this is the usual statement that energy and momentum are the conserved charges associated with symmetry in time and space.

We will give a more systematic discussion in the context of field theory, using the Noether procedure, in a few lectures.

The discussion of the procedure, which we'll come to a little bit later, will allow you to express these P's and the J's in terms of spatial integrals of the stress-energy tensor.

You can understand exactly how these are related to the stress-energy tensor. You can read about that in Chapter 7.

1.6 Microcausality and the free scalar field

So the point of this little discussion is just to explain what it means for $\hat{\phi}$ to be tied to the spacetime point x. Let me just reiterate: it's this equation here that says the Poincaré symmetry transformation, which is realized by the unitary operator U, should act as $U\hat{\phi}(x)U^{-1} = \hat{\phi}(\Lambda x + a)$, the field at the transformed spacetime point.

Okay, that's that. But so far I haven't said anything about locality, so that's going to be the key. A key property is going to be microcausality.

In this particular example, we're going to demand that $\hat{\phi}(x)$ and $\hat{\phi}(y)$ commute, where x and y are two spacetime points, whenever x and y are spacelike separated. And we'll always use the convention that $(x-y)^2$ is the Lorentzian inner product. It is defined to be minus $(x^0-y^0)^2$, the square of the time difference, plus $(\mathbf{x}-\mathbf{y})^2$, the square of the spatial difference.

So, i.e., this is positive.

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \quad \text{whenever} \quad (x-y)^2 \equiv -(x^0-y^0)^2 + (\mathbf{x}-\mathbf{y})^2 > 0. \tag{1.15}$$

Okay. Of course, as I said, I'm always working in units where the speed of light is set to one. Okay.

So that's microcausality, at least applied to this local field operator $\hat{\phi}$ we're talking about here.

A student asks whether the existence of the local field, or microcausality, is obvious here. No, it's not obvious at the moment, but we're back at the requirement of the existence of an operator $\hat{\phi}$ tied to a spacetime point x in this way. It's a nontrivial constraint on the quantum system.

Okay, so you can ask whether this applies to our system of free particles. Intuitively it should, because we have non-interacting particles, and the particles themselves propagate with the relativistic dispersion relation, so the system should be causal.

But can you actually write down these local field operators in that theory?

Yeah. What about our model, the very simple quantum-mechanical model of free relativistic particles?

So, for example, I have already written down the Hamiltonian:

$$H = \int d^{D-1}\mathbf{p}\, \sqrt{\mathbf{p}^2+m^2}\, a^+_{\mathbf{p}} a_{\mathbf{p}}. \tag{1.16}$$

You can also write the momentum with a hat to emphasize that this is an operator. We also write the capital P, the momentum operator, in a similar way:

$$\mathbf{P} = \int d^{D-1}\mathbf{p}\, \mathbf{p}\, a^+_{\mathbf{p}} a_{\mathbf{p}}. \tag{1.17}$$

I hope you can see the difference in my handwriting between the lower-case p, which is an integration variable, versus the capital P, which is the operator. All right, so the lower-case p is the variable we're integrating over. And with a μ index, this is just the vector $\hat{P}^\mu = (H, \mathbf{P})$, the energy-momentum operator.

$$\hat{P}^\mu = (H, \mathbf{P}). \tag{1.18}$$

So these are some of the generators among our symmetry generators of this model. We can also try to write the J's. They are a little bit more complicated. I won't write that right now; it will become clear how to construct that later.

But the key question you might ask is: what the heck is $\widehat{\phi}(x)$ in this model? Does it even exist?

That is, does an operator $\widehat{\phi}(x)$ exist that satisfies all of the nice properties we just demanded?

The answer is yes, but it's not totally obvious. We'll have a different way to understand this later, but for now I'll just write down the answer.

It does exist. It's not unique, but here's one. Remember the $a_{\mathbf{p}}$'s.

There's some normalization. Basically, as mentioned, it's not too important. But this is important: $\omega_{\mathbf{p}}$ is the positive-energy relativistic dispersion relation for a particle of mass m.

$$\hat{\phi}(x) = \int \frac{d^{D-1}\mathbf{p}}{\sqrt{(2\pi)^{D-1} 2\omega_{\mathbf{p}}}} \left(a_{\mathbf{p}} e^{ip\cdot x} + a_{\mathbf{p}}^{+} e^{-ip\cdot x} \right). \tag{1.19}$$

$$p \cdot x \equiv \mathbf{p} \cdot \mathbf{x} - \omega_{\mathbf{p}} x^0, \qquad \omega_{\mathbf{p}} \equiv \sqrt{\mathbf{p}^2 + m^2}. \tag{1.20}$$

And with the $a_{\mathbf{p}}$'s here, the field operator is the momentum integral written above, with $p \cdot x = \mathbf{p} \cdot \mathbf{x} - \omega_{\mathbf{p}} x^0$.

And I'm integrating over $\mathbf{p}$. This $\omega_{\mathbf{p}}$ is just the energy according to the dispersion relation, right, by inspection.

Now, I don't have time to explain this right now. I will invite you to think on this for yourself. I will give some explanation next time, but we'll understand this much more systematically later.

That is to say, this construction will obey all the nice properties that we just imagined. It's going to transform under Poincaré transformations in the right way and, importantly, it's going to obey microcausality.

Check micro causality.

$$[\hat{\phi}(x), \hat{\phi}(y)] = \int \frac{d^{D-1}\mathbf{p}}{(2\pi)^{D-1} 2\omega_{\mathbf{p}}} \cdot \left[e^{ip\cdot(x-y)} - e^{-ip\cdot(x-y)} \right]. \tag{1.21}$$

$$[\hat{\phi}(x), \hat{\phi}(y)] = \int \underbrace{\frac{d^D p}{(2\pi)^{D-1}} \theta(p^0) \delta(p^2 + m^2)}_{\substack{\text{invariant for} \\ \Lambda \in SO^+(1,D-1)}} \underbrace{\left[e^{ip\cdot(x-y)} - e^{-ip\cdot(x-y)} \right]}_{\substack{\text{scalar under simultaneous} \\ p \to \Lambda p,\ (x-y) \to \Lambda(x-y)}}. \tag{1.22}$$

Under a proper orthochronous Lorentz transformation, both p and $z = x - y$ transform; changing the integration variable leaves the integral invariant. For spacelike z, choose a Lorentz frame with $z^0 = 0$ and $\mathbf{z} \neq 0$. The integrand is odd under $\mathbf{p} \to -\mathbf{p}$, so the integral vanishes.

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \qquad \text{for} \qquad (x-y)^2 > 0. \tag{1.23}$$

The free field is also required to obey the covariance property below. Its direct verification is deferred until the systematic construction.

$$U(\Lambda, a)\, \hat{\phi}(x) \left(U(\Lambda, a) \right)^{-1} = \hat{\phi}(\Lambda x + a). \tag{1.24}$$

1.7 Postulates and the next formulation

The handwritten source postulates that the same properties hold for a system of interacting relativistic particles:

- a Hilbert space $\mathcal{H}$;
- Poincaré symmetry $U(\Lambda, a)$;
- a Poincaré-invariant vacuum $|\Omega\rangle$;
- local "field" operators $\hat{\phi}(x)$ obeying Poincaré covariance and microcausality.

An interacting system that had similar properties, that had this kind of microcausality, as you see, would not be very easy to construct in the Hamiltonian formalism, for the basic reason that the Hamiltonian formalism singles out time versus space. Poincaré symmetry is not very manifest.

For example, here, you know, we're often writing integrals over spatial momentum. That doesn't look obviously Lorentz covariant, even though the underlying physics is Lorentz invariant. You'll be able to construct operators, but it's not trivial to check that the system actually has that symmetry.

So it will be useful to have a formulation in which Poincaré symmetry is manifest, and that cannot be done through the Hamiltonian, and that's why the Lagrangian formalism will be useful. Next time we'll discuss how to formulate quantum mechanics using the Lagrangian language.

Final question

Okay, any final questions before we end? Yes.

A student asks how the displayed field operator was constructed. For now, yes.

Because this is only a, you know, kind of general motivation. Of course this is the correct answer. You can check, or you might check, that it works, but we'll explain how to systematically construct this thing.

Editorial note. Problem Set 1 on source physical pages 15–19 is deferred for later integration into the exercises appendix.